Case *Study*

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Context

Target is a globally renowned brand and a prominent retailer in the United States. Target makes itself a preferred shopping destination by offering outstanding value, inspiration, innovation and an exceptional guest experience that no other retailer can deliver.

This particular business case focuses on the operations of Target in Brazil and provides insightful information about 100,000 orders placed between 2016 and 2018. The dataset offers a comprehensive view of various dimensions including the order status, price, payment and freight performance, customer location, product attributes, and customer reviews.

By analyzing this extensive dataset, it becomes possible to gain valuable insights into Target's operations in Brazil. The information can shed light on various aspects of the business, such as order processing, pricing strategies, payment and shipping efficiency, customer demographics, product characteristics, and customer satisfaction levels.

Dataset:

https://drive.google.com/drive/folders/1TGEc66YKbD443nsIRi1bWgVd238gJCnb

The data is available in 8 csv files:

1. customers.csv
2. sellers.csv
3. order_items.csv
4. geolocation.csv
5. payments.csv
6. reviews.csv
7. orders.csv
8. products.csv

The column description for these csv files is given below.

customers.csv

Features	Description
customer_id	ID of the consumer who made the purchase
customer_unique_id	Unique ID of the consumer
customer_zip_code_prefix	Zip Code of consumer's location
customer_city	Name of the City from where order is made
customer_state	State Code from where order is made (Eg. são paulo - SP)

sellers.csv

Features	Description
seller_id	Unique ID of the seller registered
seller_zip_code_prefix	Zip Code of the seller's location
seller_city	Name of the City of the seller
seller_state	State Code (Eg. são paulo - SP)

order_items.csv

Features	Description
order_id	A Unique ID of order made by the consumers
order_item_id	A Unique ID given to each item ordered in the order
product_id	A Unique ID given to each product available on the site
seller_id	Unique ID of the seller registered in Target
shipping_limit_date	The date before which the ordered product must be shipped
price	Actual price of the products ordered
freight_value	Price rate at which a product is delivered from one point to another

geolocations.csv

Features	Description
geolocation_zip_code_prefix	First 5 digits of Zip Code
geolocation_lat	Latitude
geolocation_lng	Longitude
geolocation_city	City
geolocation_state	State

payments.csv

Features	Description
order_id	A Unique ID of order made by the consumers
payment_sequential	Sequences of the payments made in case of EMI
payment_type	Mode of payment used (Eg. Credit Card)
payment_installments	Number of installments in case of EMI purchase
payment_value	Total amount paid for the purchase order

orders.csv

Features	Description
order_id	A Unique ID of order made by the consumers
customer_id	ID of the consumer who made the purchase
order_status	Status of the order made i.e. delivered, shipped, etc.
order_purchase_timestamp	Timestamp of the purchase
order_delivered_carrier_date	Delivery date at which carrier made the delivery
order_delivered_customer_date	Date at which customer got the product
order_estimated_delivery_date	Estimated delivery date of the products

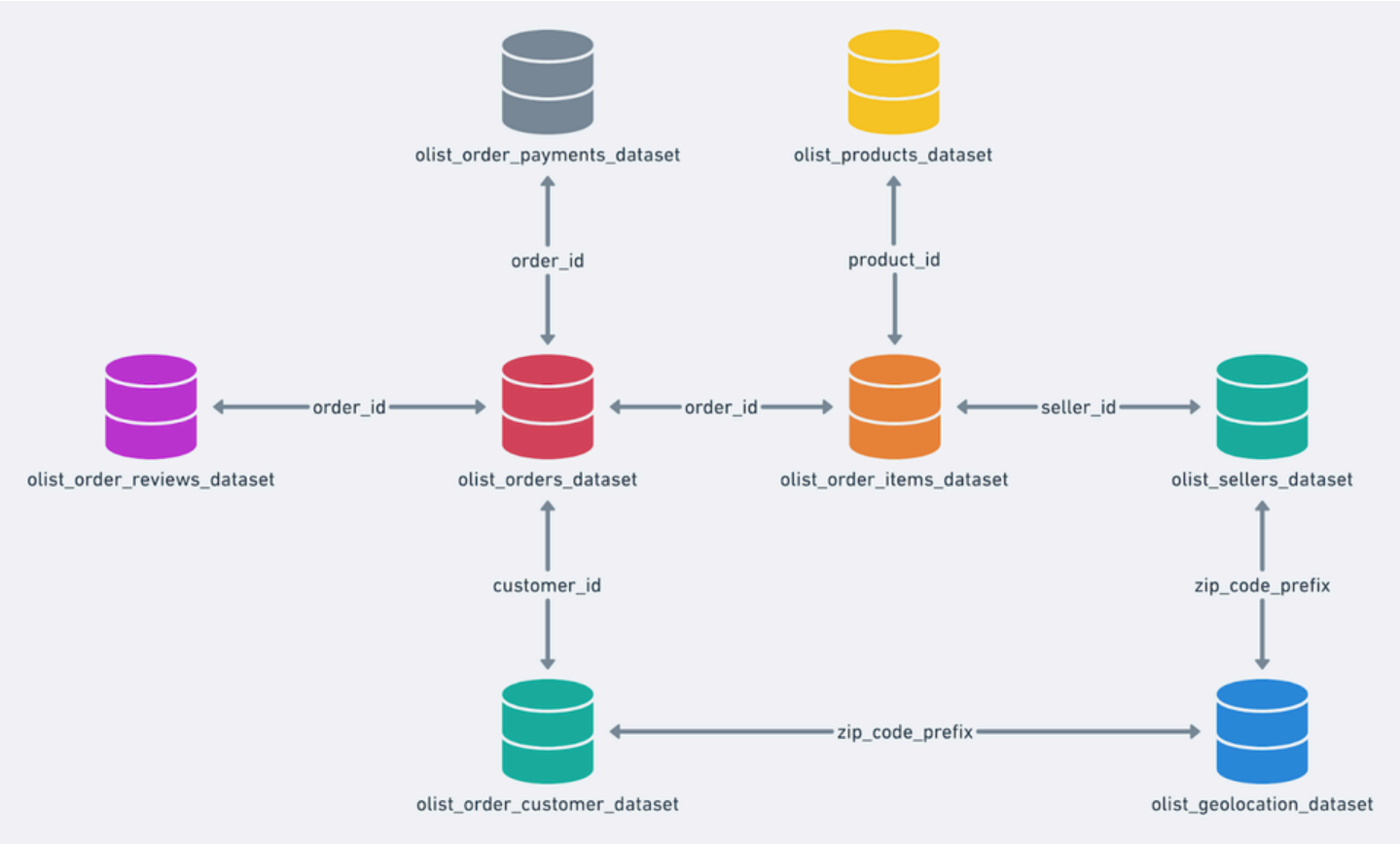
The **reviews.csv** contain following features:

Features	Description
review_id	ID of the review given on the product ordered by the order id
order_id	A Unique ID of order made by the consumers
review_score	Review score given by the customer for each order on a scale of 1-5
review_comment_title	Title of the review
review_comment_message	Review comments posted by the consumer for each order
review_creation_date	Timestamp of the review when it is created
review_answer_timestamp	Timestamp of the review answered

products.csv

Features	Description
product_id	A Unique identifier for the proposed project.
product_category_name	Name of the product category
product_name_lenght	Length of the string which specifies the name given to the products ordered
product_description_lenght	Length of the description written for each product ordered on the site
product_photos_qty	Number of photos of each product ordered available on the shopping portal
product_weight_g	Weight of the products ordered in grams
product_length_cm	Length of the products ordered in centimeters
product_height_cm	Height of the products ordered in centimeters
product_width_cm	Width of the product ordered in centimeters

Dataset schema:



Problem Statement:

Analyze the given dataset to extract valuable insights and provide actionable recommendations.

1.Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1.Data type of all columns in the "customers" table.

Query:

```
SELECT table_name,column_name,data_type
FROM target.INFORMATION_SCHEMA.COLUMNS
where table_name = 'customers';
```

Query results

Job information | Results | Visualisation | JSON | Execution de

Row	table_name	column_name	data_type
1	customers	customer_id	STRING
2	customers	customer_unique_id	STRING
3	customers	customer_zip_code_prefix	INT64
4	customers	customer_city	STRING
5	customers	customer_state	STRING

Insights:

- [ ] There are 5 columns -4 string and one integer(zip code)
- [ ] The 3 customer of the columns are related to location

Recommendations:

- [ ] adding 4 null-able fields called :
  - [ ] “name”(personalizing)
  - [ ] ”phone number”(staying on top of their mind(phone push notifications))
  - [ ] ”date-of-birth”(putting them in better basket together)
  - [ ] ”gender” (customization)

2.Get the time range between which the orders were placed.

Query:

```
select *, round(date_diff( a.maximum_date,a.minimun_date, day)/365,1) as
time_frame_years from
(SELECT min(order_purchase_timestamp) minimun_date,
max(order_purchase_timestamp) maximum_date
FROM `target-atchu.target.orders`)a
```

Row	minimun_date	maximum_date	time_frame_years
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC	2.1

Insights:

- [ ] The dataset has tuples from 04-sep-2016 till 17-oct-2018 (almost 2 years)

3.Count the Cities & States of customers who ordered during the given period.

Query:

```
select
count(distinct customer_city) cnt_city,
count(distinct customer_state) cnt_state
from target-atchu.target.customers c
where exists (select 1 from target-atchu.target.orders where customer_id=c.customer_id)
```

Note: we can filter order_purchase_timestamp between those two given ranges but are not necessary because we are considering the entire time frame!

Query results

Job information | Results | Visualisation

Row	cnt_city	cnt_state
1	4119	27

Insights:

- [ ] The store operates from 4119 distinct cities and across 27 states in Brazil in between the period of 04-sep-2016 till 17-oct-2018 (almost 2 years)

2.In-depth Exploration:

1.Is there a growing trend in the no. of orders placed over the past years?

Query:

Initial analysis shows there has been a clear growth from year to year.

```
select extract(year from order_purchase_timestamp) year,
count(order_id) no_of_orders
from target-atchu.target.orders
group by year
order by year;
```

Row	year	no_of_orders
1	2016	329
2	2017	45101
3	2018	54011

But on closer observation few alarming facts have been uncovered

```
select *, round((no_of_orders-lag(no_of_orders) over(order by
year,month))/lag(no_of_orders) over(order by year,month)*100,2) percentage_change
from (select extract(year from order_purchase_timestamp) year,
extract(month from order_purchase_timestamp) month,
count(order_id) no_of_orders
from target-atchu.target.orders
group by year,month ) a
order by year,month
```

Row	year	month	no_of_orders	percentage_chan...
1	2016	9	4	null
2	2016	10	324	8000.0
3	2016	12	1	-99.69
4	2017	1	800	79900.0
5	2017	2	1780	122.5
6	2017	3	2682	50.67
23	2018	8	6512	3.5
24	2018	9	16	-99.75
25	2018	10	4	-75.0

Insights:

- The **sharp increase** in the **initial months** show that the store was a **huge success**.
- the store **gradually stabilized** with a consistent customer base with occasional seasonal spikes**.**
- A gradual decline has been observed from November 2017 onward.
- A **sharp decline** has been observed in the **September and October of the 2018.** We need to confirm if that is the actual data or if there is a data loss during extraction.

Recommendations:

- Confirm if it is the actual data or if there has been a loos in data when extracting.
- If it is indeed the actual data, need to check the steep downward spike in the months of September and October 2018.

2.Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Query:

Initial analysis shows there has been a clear spike in months 8,5,7.

```
select extract(month from order_purchase_timestamp) month,
count(order_id) no_of_orders
from target-atchu.target.orders
group by month order by  no_of_orders desc;
```

Job information | Results | Visualisat

Row	month	no_of_orders
1	8	10843
2	5	10573
3	7	10318
4	3	9893
5	6	9412
6	4	9343
7	2	8508
8	1	8069

But on closer observation we understood the below insights.

```
select month,round(avg(avg_per_month),2) monthly_aggregate_Avg from
(select distinct year,month,
count(order_id) over(partition by year,month )/count(order_id) over(partition by year)*100
avg_per_month from
```

```
(select extract(year from order_purchase_timestamp) year,
extract(month from order_purchase_timestamp) month,
order_id from target-atchu.target.orders
where order_purchase_timestamp between '2017-02-01' and '2018-10-31' ) a)b
group by month order by monthly_aggregate_Avg desc;
```

Note: We consider the percentage of the monthly data in the specified time frame (after the store has actually stabilised) to ensure the results are not skewed. Had we considered the count of months in entire dataset we would have assumed that months 8,5,7 showed promise which would have been misleading.

Row	month	monthly_aggrega...
1	11	17.03
2	1	13.46
3	12	12.81
4	8	10.92
5	5	10.54
6	7	10.37
7	3	9.7
8	6	9.37
9	4	9.14
10	2	8.24

Insights:

- [ ] November is the month with highest orders probably owing to the Brazilian carnival, Halloween, pre-Christmas frenzy.
- [ ] January is the second highest sales (New year resolutions possibly?)
- [ ] It's noteworthy that the store could maintain the footfall/spike (with a marginal loss) for almost 6 months.
- [ ] also had we considered the actual delivered orders, it may have shown a different story too.

Recommendations:

- [ ] Need to do heavy bulk (basket should include medium performers along with instant hits) offers/promotions deals in the holiday season to improve over all orders.
- [ ] Need to offer attractive MBC in the following seasons to retain the footfall.
- [ ] Need to do clearance sales in the months with lower spike to encourage improvement in footfall.

3.During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

Query:

```
select
case when extract(hour from order_purchase_timestamp) between 0 and 6 then 'Dawn'
when extract(hour from order_purchase_timestamp) between 7 and 12 then 'Morning'
when extract(hour from order_purchase_timestamp) between 13 and 18 then 'Afternoon'
        Else 'Night' end order_frame,count(order_id) no_of_orders
from target-atchu.target.orders
group by order_frame
order by  no_of_orders desc;
```

Row	order_frame	no_of_orders
1	Afternoon	38135
2	Night	28331
3	Morning	27733
4	Dawn	5242

Insights:

- [ ] most orders are ordered in the afternoons.
- [ ] night and morning are almost the same.
- [ ] dawn has the least traffic.

Recommendations:

- [ ] Plan most time related offers in the afternoons.
- [ ] Plan system daily refresh in dawn.

3.Evolution of E-commerce orders in the Brazil region:

1.Get the month on month no. of orders placed in each state.

Query:

```
select *,dense_rank() over(partition by yymm order by no_orders_per_state desc)
state_contribution_rank
from (select distinct format_date('%Y-%m',order_purchase_timestamp) YYMM,
c.customer_state, count(order_id) over(partition by format_date('%Y-
%m',order_purchase_timestamp), c.customer_state) no_orders_per_state,
count(distinct customer_state) over(partition by format_date('%Y-
%m',order_purchase_timestamp)) no_states_penetrated_per_month
from target-atchu.target.orders o join target-atchu.target.customers c
using(customer_id))a
order by YYMM,state_contribution_rank
```

Row	YYMM	customer_state	no_orders_per_state	no_states...	state_contributi...
1	2016-09	SP	2	3	1
2	2016-09	RR	1	3	2
3	2016-09	RS	1	3	2
4	2016-10	SP	113	21	1
5	2016-10	RJ	56	21	2
49	2017-01	MS	1	24	18
50	2017-02	SP	654	27	1
51	2017-02	MG	259	27	2
558	2018-08	AP	2	26	25
559	2018-09	SP	8	4	1
560	2018-09	MG	4	4	2

Insights:

- The store was first launched in states SP,RR,RS.
- It was launched in
  - 21 states in the oct-16
  - 24 states in the jan-17
  - 27 states in feb-17
- There have been occasional store closures(probably maintenance or store enhancements) in some months.
- The state SP has a consistent presence with growing spike almost every month.

- [ ] The sep/oct 18 data strongly assists the data extraction error assumption.

Recommendations:

- [ ] Nurture/incentivise the state 'SP' which has the most reception by giving better offers.
- [ ] Analyse and implement the success of 'SP' across all states.

2.How are the customers distributed across all the states?

Query:

```
select customer_state,count(customer_id)cnt
from target-atchu.target.customers c
group by customer_state order by cnt desc;
```

Row	customer_state	cnt
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020

Insights:

- [ ] The store SP(first launch state) has the maximum customer base.
- [ ] Even the second greatest state doesn't even have half of the first state.
- [ ] The orders are proportional to the customer base.

Recommendations:

- [ ] Need to improve the customer base across all cities maybe through referral bonus, roadshows, competitor analysis.

4.Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1.Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).
You can use the "payment_value" column in the payments table to get the cost of orders.

Query:

```
select *,round((order_price-prev_month_price)/prev_month_price*100,2)
percentage_change from (
select *,lag(order_price) over(order by year,month) prev_month_price from (
select extract(year from order_purchase_timestamp) year,
extract(month from order_purchase_timestamp) month, count(distinct order_id)
no_of_orders,
round(sum(payment_value),2) order_price
from target-atchu.target.orders join target-atchu.target.payments using (order_id)
where order_purchase_timestamp between '2017-01-01' and '2018-08-31'
group by  year,month)a)b
order by year,month
```

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	year	month	no_of_orders	order_price	prev_month_price	percentage_chan...
1	2017	1	800	138488.04	null	null
2	2017	2	1780	291908.01	138488.04	110.78
3	2017	3	2682	449863.6	291908.01	54.11
4	2017	4	2404	417788.03	449863.6	-7.13
5	2017	5	3700	592918.82	417788.03	41.92
6	2017	6	3245	511276.38	592918.82	-13.77
7	2017	7	4026	592382.92	511276.38	15.86
8	2017	8	4331	674396.32	592382.92	13.84
9	2017	9	4285	727762.45	674396.32	7.91
10	2017	10	4631	779677.88	727762.45	7.13

Insights:

- [ ] The order costs has grown proportional to the no of order placed.
- [ ] This also captures the seasonality that we saw earlier (the raise in peak and immediate dip)
- [ ] Also it captures the raise in payment in the month of Jan 2018 wherein the payment

Recommendations:

- [ ] Considering from February 2017 will flatten the spike.

2.Calculate the Total & Average value of order price for each state.

Query:

```
select customer_state, round(sum(tp),2) total_payment, round(avg(apv),2) avg_payment
from target-atchu.target.orders o
join (select order_id,sum(payment_value) tp, avg(payment_value) apv from target-atchu.target.payments group by order_id) p using (order_id)
join target-atchu.target.customers c using(customer_id)
group by customer_state order by total_payment desc
```

Row	customer_state	total_payment	avg_payment
1	SP	5998226.96	141.39
2	RJ	2144379.69	163.94
3	MG	1872257.26	158.41
4	RS	890898.54	160.21
5	PR	811156.38	158.55
6	SC	623086.43	168.69
7	BA	616645.82	178.45
8	DF	355141.08	163.24
9	GO	350092.31	170.45
10	ES	325967.55	158.02

Insights:

- This adds to the strong fact that the state SP has been the strong customer base and hence the strongest market though having a lower spending avg, even though the total payment is more than the double of the second highest one.
- Though MG has a stronger customer base it has lower payment count than RJ
- All the other state has a lower customer base but has a stronger avg customer payment.

Recommendations:

- Plan the offers accordingly
  - In SP target carters low -mid premium customers who feel this is their normal retail.
  - In most of the other states this retail feels like mid premium or premium(GO has avg payment 233) customer brand.
- Its important to penetrate all the strata of customers and ensure we feel relevant to them, so the offers and quality of products should go hand in hand.

3.Calculate the Total & Average value of order freight for each state.

Query:

We checked the freight per state

```
select customer_state, round(sum(tp),2) total_payment, round(avg(apv),2) avg_payment
from target-atchu.target.orders o
join (select order_id,sum(freight_value) tp, avg(freight_value) apv from target-atchu.target.order_items group by order_id) p using (order_id)
join target-atchu.target.customers c using(customer_id)
group by customer_state order by total_payment desc
```

Row	customer_state	total_payment	avg_payment
1	SP	718723.07	15.3
2	RJ	305589.31	21.15
3	MG	270853.46	20.8
4	RS	135522.74	21.97
5	PR	117851.68	20.45
6	BA	100156.68	26.26
7	SC	89660.26	21.78
8	PE	59449.66	32.81
9	GO	53114.98	23.11
10	DF	50625.5	21.34

And on comparison with payment value we understood the below insights.

```
select customer_state,round(x.tp,2) total_payment,round(x.tfv,2) total_freight_value,round((x.tp-x.tfv),2) actual_profit_value, round(x.ap,2) avg_payment,round(x.afv,2) avg_freight_value, round(x.ap-x.afv,2) avg_spend_wo_freight from
(select customer_state, round(sum(tfv),2) tfv,avg(afv) afv, sum(tp) tp,avg(ap) ap
from target-atchu.target.orders o
join (select order_id,sum(freight_value) tfv,avg(freight_value) afv from target-atchu.target.order_items group by order_id) p using (order_id)
join(select order_id,sum(payment_value) tp,avg(payment_value) ap from `target-atchu.target.payments` group by order_id) t using (order_id)
join target-atchu.target.customers c using(customer_id)
group by customer_state)x  order by total_freight_value desc;
```

Row	customer_state	total_profit	total_freight_value	actual_profit_value	avg_payment	avg_freight_value	avg_spend_wo_fr...
1	SP	5922248.07	718714.58	5203533.49	140.84	15.3	125.54
2	RJ	2130002.87	305589.31	1824413.56	163.99	21.15	142.84
3	MG	1856547.43	270853.46	1585693.97	158.41	20.8	137.61
4	RS	886156.37	135522.74	750633.63	160.36	21.97	138.39
5	PR	801146.18	117851.68	683294.5	158.06	20.45	137.61
6	BA	611639.46	100156.68	511482.78	178.16	26.26	151.9
7	SC	610294.22	89660.26	520633.96	166.33	21.78	144.55
8	PE	322340.23	59449.66	262890.57	192.53	32.81	159.72
9	GO	347788.8	53114.98	294673.82	170.4	23.11	147.29
10	DF	353252.26	50625.5	302626.76	163.52	21.34	142.18
11	ES	324826.63	49764.6	275062.03	158.14	22.18	135.96

Insights:

- The average freight cost is the lowest in the state SP, probably one of the reasoons why the state is performing the best.
- The “premium” customer states have a higher freight_value but even if we remove the freight charges their avg spend is in the premium category.

Recommendations:

- Need to reduce freight charges as much as possible (unless there is a demand for the same seller- Turkish pistachios are low rang in source and premium when sourced).

5.Analysis based on sales, freight and delivery time

1.Find the no. of days taken to deliver each order from the order's purchase date as delivery time. Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

Query:

```
select order_id,
date_diff(order_delivered_customer_date,order_purchase_timestamp,day)
time_to_deliver_days,
date_diff(order_delivered_customer_date,order_estimated_delivery_date,day)
diff_estimated_delivery
from `target-atchu.target.orders`
where order_status='delivered';
```

Row	order_id	time_to_deliver_days	diff_estimated_delivery
1	bfbd0f9bdef84302105ad712db...	54	36
2	98974b076b01553d49ee64679...	43	-6
3	c4b41c36dd589e901f6879f25a...	36	-14
4	d2292ff2201e74c5db154d1b7a...	29	-20
5	95e01270fcbae986342340010...	30	-19
6	ed8c7b1b3eb256c70ce0c7423...	44	-5
7	5cc475c7c03290048eb2e742c...	68	18
8	6b3ee7697a02619a0ace2b3f0a...	47	-2
9	3b2ca3293a7ce539ea2379d70...	43	-7
10	b2f92b2f7047cd8b35580d629d...	43	-7

Insights:

- the minimum an order took to be delivered is the same day
- The maximum an order took to be delivered 209days(almost 7 months from delivery date and unless it was a occasion specific pre-booking order like customised automobile, wedding dress or cake that is too much!)
- The earliest an order has been delivered before estimated date is 181 days before.
- The latest it has been delivered is 146 days before.

Recommendations:

- Need to map the seller closer to buyer for fastest delivery using geolocation.

2.Find out the top 5 states with the highest & lowest average freight value.

Query:

```
select if(min_fval_states<=5,customer_state,null) min_fval_states,afv,
if(max_fval_states<=5,customer_state,null) max_fval_states
from (select customer_state, afv,
dense_rank() over(order by afv desc) max_fVal_states,
dense_rank() over(order by afv) min_fval_states from
(select customer_state,round(avg(freight_value),2) afv
from `target-atchu.target.orders`
join `target-atchu.target.order_items` using(order_id)
join `target-atchu.target.customers` using(customer_id)
group by customer_state) x ) y
where max_fval_states<=5 or min_fval_states<=5
```

Row	min_fval_states	afv	max_fval_states
1	SP	15.15	null
2	PR	20.53	null
3	MG	20.63	null
4	RJ	20.96	null
5	DF	21.04	null
6	null	39.15	PI
7	null	40.07	AC
8	null	41.07	RO
9	null	42.72	PB
10	null	42.98	RR

Insights:

- The states with minimum freight charge have more orders and perform well.
- the states with more freight charges have less orders and perform comparatively poor.

Recommendations:

- Need to map the seller closer to buyer for fastest delivery using geolocation table.

3.Find out the top 5 states with the highest & lowest average delivery time.

Query:

```
select if(min_fval_states<=5,customer_state,null) min_val_states,
afv avg_time_for_delivery,
if(max_fval_states<=5,customer_state,null) max_val_states
from (select customer_state, afv,
dense_rank() over(order by afv desc) max_fVal_states,
dense_rank() over(order by afv) min_fval_states from
(select customer_state,
round(avg(date_diff(order_delivered_customer_date,order_purchase_timestamp,day)),2
) afv from `target-atchu.target.orders`
join `target-atchu.target.customers` using(customer_id)
group by customer_state) x ) y
where max_fval_states<=5 or min_fval_states<=5
```

Row	min_val_states	avg_time_for_deli...	max_val_states
1	SP	8.3	null
2	PR	11.53	null
3	MG	11.54	null
4	DF	12.51	null
5	SC	14.48	null
6	null	23.32	PA
7	null	24.04	AL
8	null	25.99	AM
9	null	26.73	AP
10	null	28.98	RR

Insights:

- The states with minimum time to deliver have more orders and perform well.
- the states with more time to deliver have less orders and perform comparatively poor.

Recommendations:

- Need to map the seller closer to buyer for fastest delivery using geolocation table.

4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

Query:

```
select if(min_fval_states<=5,customer_state,null) min_val_states,
afv est_vs_act_deliv_avg,
 if(max_fval_states<=5,customer_state,null) max_val_states from
(select customer_state,afv,dense_rank() over(order by afv desc)
max_fVal_states,dense_rank() over(order by afv) min_fval_states
from (select customer_state,
round(avg(date_diff(order_delivered_customer_date,order_estimated_delivery_date,day)),2) afv from `target-atchu.target.orders`
join `target-atchu.target.customers` using(customer_id)
where order_status='delivered'
group by customer_state) x ) y
where max_fval_states<=5 or min_fval_states<=5
```

Row	min_val_states	est_vs_act_deliv_...	max_val_states
1	AL	7.95	null
2	MA	8.77	null
3	SE	9.17	null
4	ES	9.62	null
5	BA	9.93	null
6	null	16.41	RR
7	null	18.61	AM
8	null	18.73	AP
9	null	19.13	RO
10	null	19.76	AC

Insights:

- [ ] Estimation of delivery dates is way off. The order is delivered way before it was actually estimated to be delivered.

Recommendations:

- [ ] Overestimating the delivery date throws the customer off and they may not want to order.

6.Analysis based on the payments

1.Find the month on month no. of orders placed using different payment types.

Query:

```
select extract(year from order_purchase_timestamp) year,
extract(month from order_purchase_timestamp) month,
payment_type, round(sum(payment_value),2) total_payment
from `target-atchu.target.orders` o
join `target-atchu.target.payments` p using(order_id)
where payment_type!='not_defined'
group by year,month,payment_type
order by year,month,total_payment desc
```

Row	year	month	payment_type	total_payment
1	2016	9	credit_card	252.24
2	2016	10	credit_card	48290.62
3	2016	10	UPI	9679.06
4	2016	10	voucher	879.07
5	2016	10	debit_card	241.73
6	2016	12	credit_card	19.62
7	2017	1	credit_card	109615.68
8	2017	1	UPI	24074.43
9	2017	1	voucher	4054.4
10	2017	1	debit_card	743.53

Insights:

- [ ] There are 4 payment types (credit, upi,debit,voucher)
- [ ] Credit (installment based) has been the most successful.

Recommendations:

- [ ] Try to tie up with no-cost EMI based financing models.

2.Find the no. of orders placed on the basis of the payment installments that have been paid.

Query:

```
select count(distinct if( payment_sequential = payment_installments,order_id,null
))paid_installment_orders,
count(distinct order_id)total_installment_orders from `target-atchu.target.payments`
where payment_type in('credit_card','debit_card');
```

Row	paid_installment_orders	total_installment_orders
1	26872	78032

Insights:

- [ ] Only around 25% of orders have been repaid.

Insights and recommendations

- Target started operations in Brazil in sep 2016 in SP state and took almost until jan 2017 to stabilise.
- Its stores grew from 4(sep'16) to 21(oct'16) to 24(jan'17) and operates across 27(feb'17) states right now.
- The stores show seasonality in the footfall with the Brazilian carnival season showing the maximum footfall and the january being the second.
- The stores that have least freight_Value and offer fastest deliveries are performing better and have better customer base owing to the fact that the customers can get the products easily and affordably.
- Customers prefer to by in installments.

- Utilise the seasonality and create basket promotions to carter the needs and create "false" seasonality by creating discounts(clearances) in the slow performing months.
- Find sellers closer to the buyers to reduce freight_cost, delivery time to increase state performances.
- create tie ups with financial institutions to offer no cost EMIs and invest wisely to balance the waived interest.